

Bearing Failure Analysis

Bearing failure analysis

Overview:

- Load path patterns
- Failure mode classification
- How to secure evidence
- Hands-on exercises

Bearing failure analysis

The best way to become expert is to examine damaged bearings. Evidence will emerge to allow root cause analysis and hence corrective action. Many of ball and roller bearings never attain their calculated life expectancy. .

The calculated life expectancy of any bearing is based on four assumptions:

1. Good lubrication in proper quantity will always be available to the bearing.
2. The bearing will be mounted without damage.
3. Dimensions of parts related to the bearing will be correct.
4. There are no defects inherent in the bearing.

Bearing failure analysis

Even if all the four conditions are met, the bearing may still fail due to fatigue of bearing material.

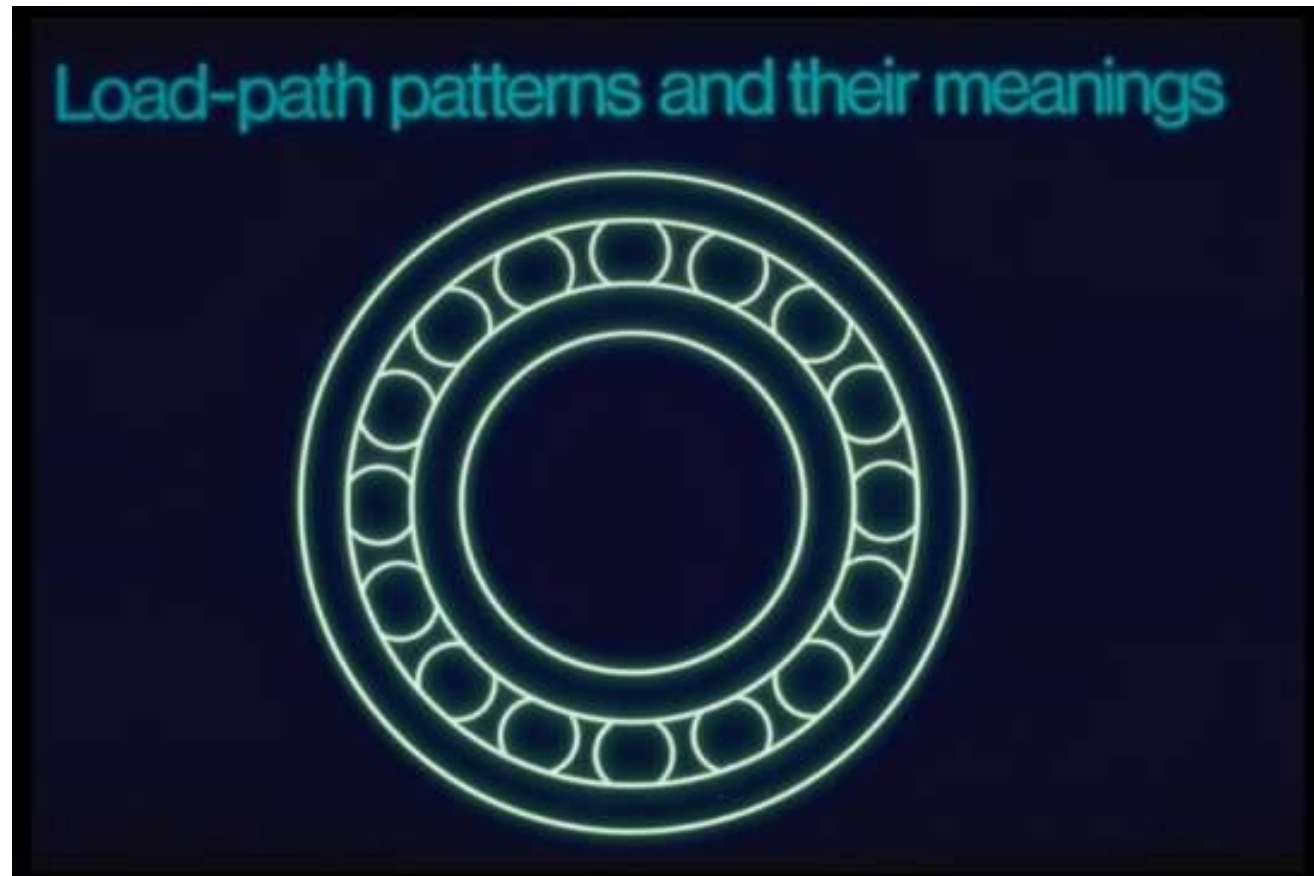
Three major classifications of premature spalling are lubrication, mechanical damage, and material defects.

Bearing failure analysis

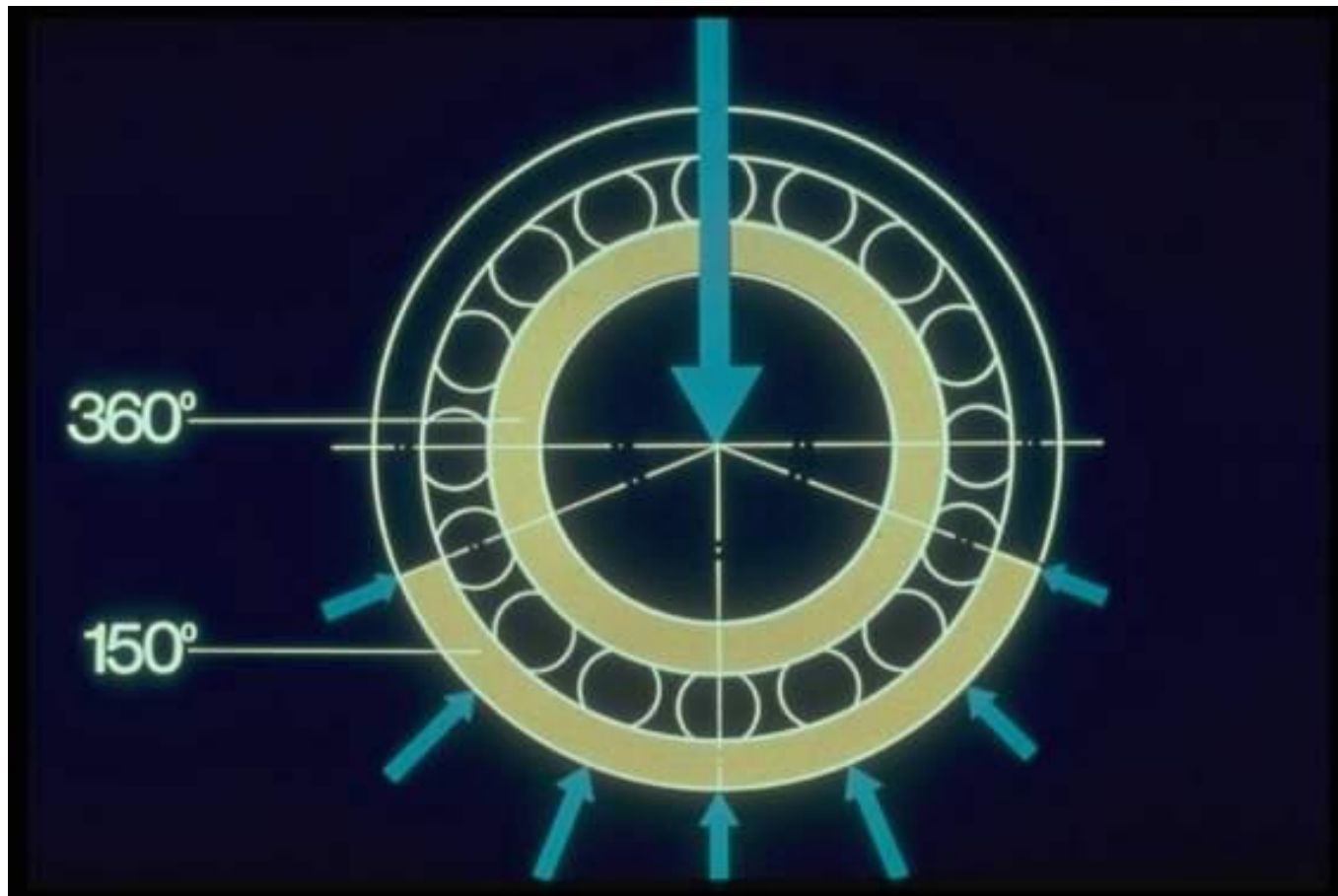
Most bearing failures can be attributed to one or more of the following causes:

1. Defective bearing seats on shafts and in housings.
2. Misalignment.
3. Faulty mounting practice
4. Incorrect shaft and housing fits.
5. Inadequate lubrication.
6. Ineffective sealing.
7. Vibration while the bearing is not rotating
8. Passage of electric current through the bearing.
9. Transportation, storage and handling.

Load paths

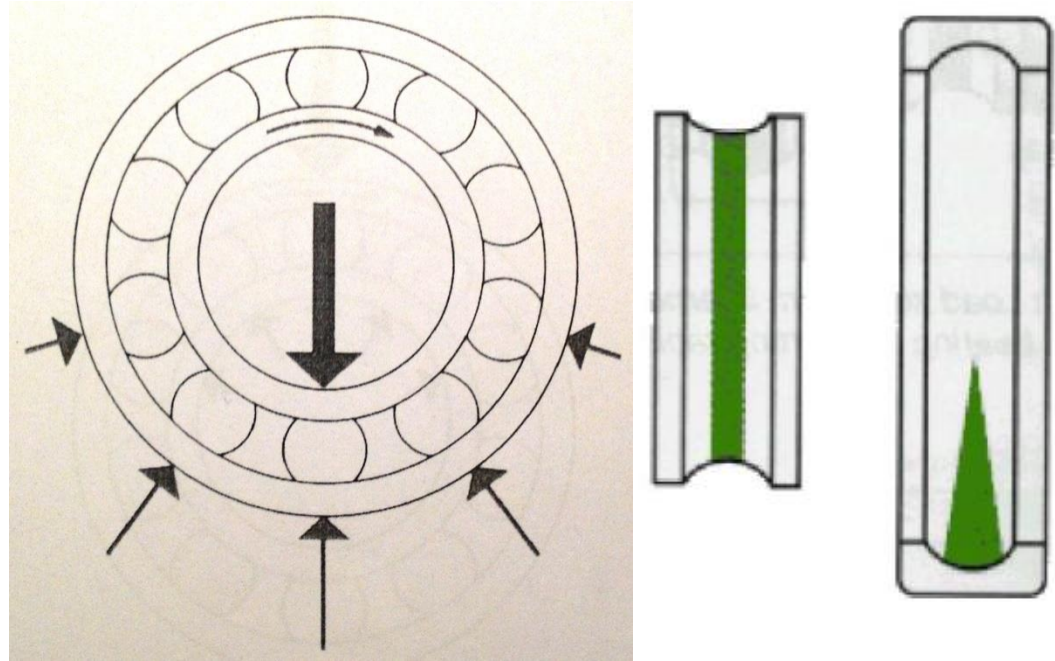


Normal radial load zones...



Normal radial load zones...

Stationary outer ring
Rotating inner ring
Constant unidirectional loads

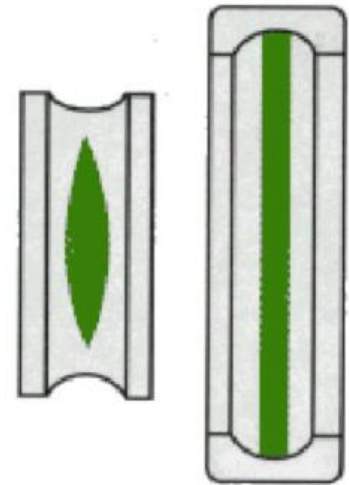
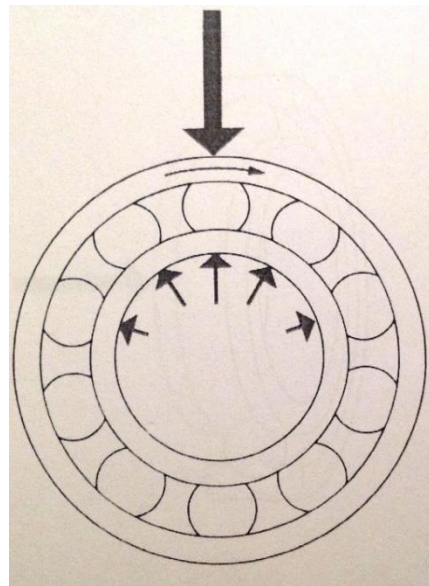


Normal radial load zones

Stationary inner ring

Rotating outer ring

Constant unidirectional load

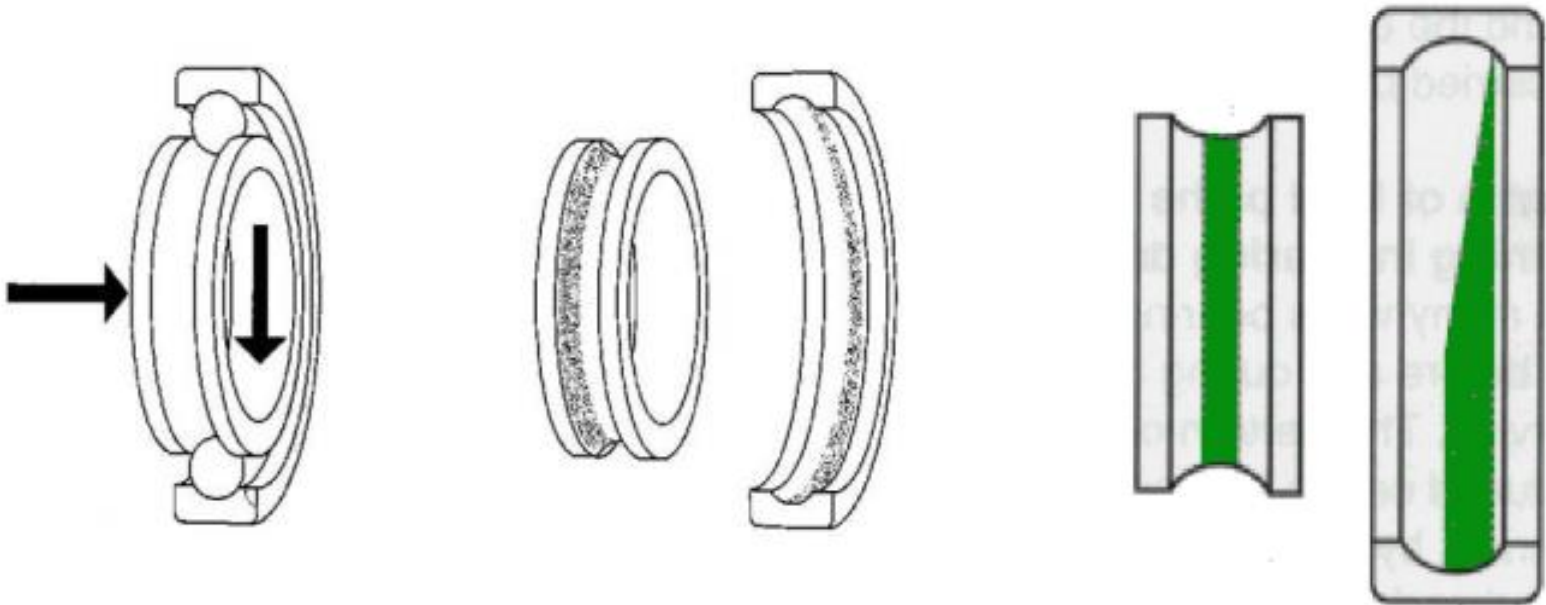


Combined (radial and axial) load

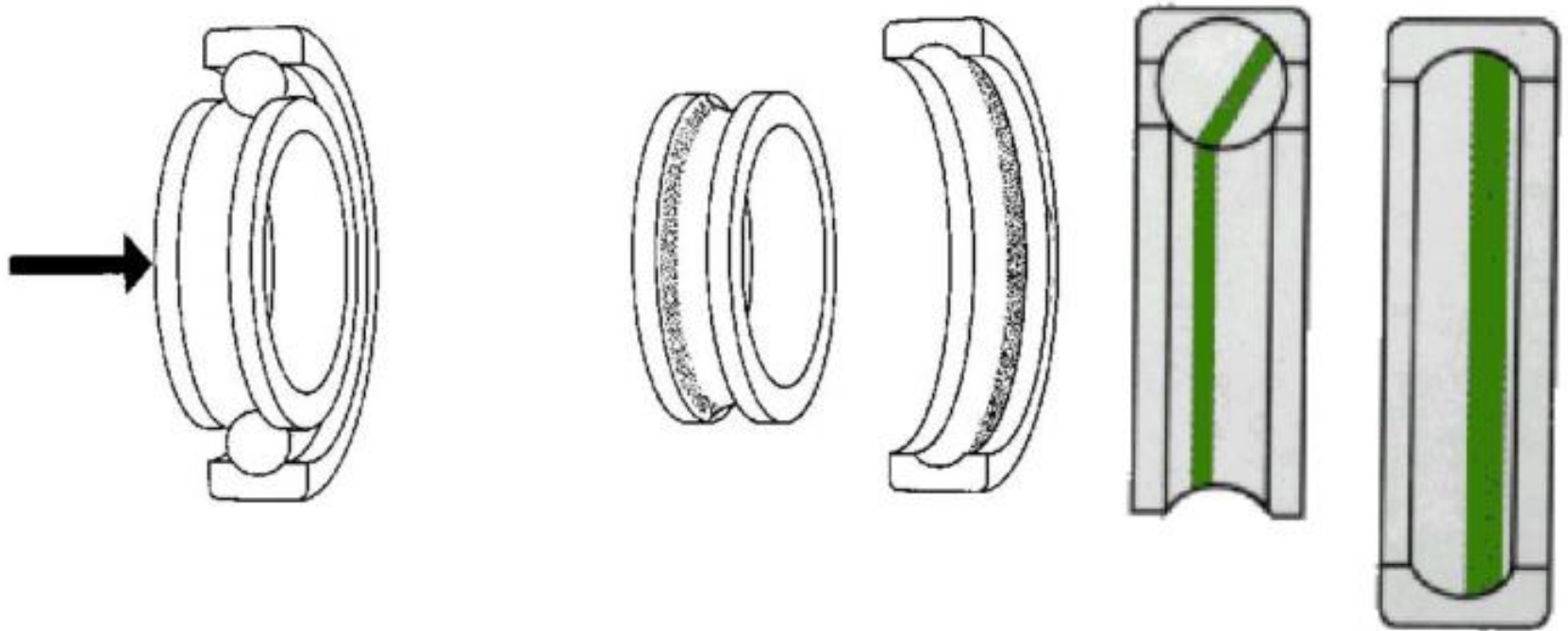
Stationary outer ring

Rotating inner ring

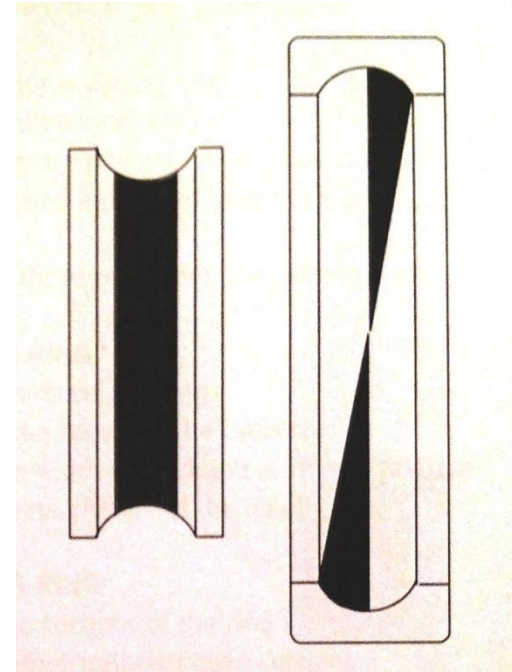
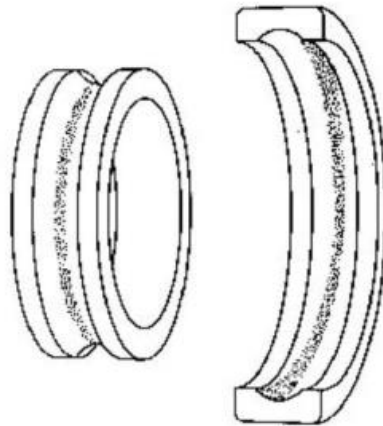
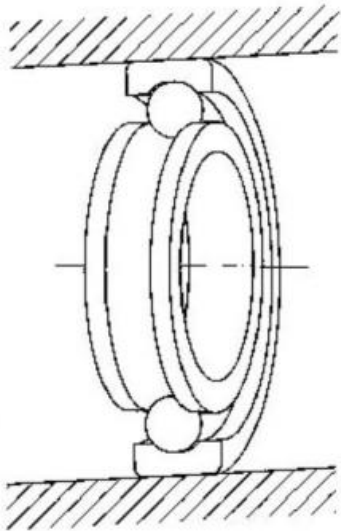
Constant unidirectional radial and axial loads



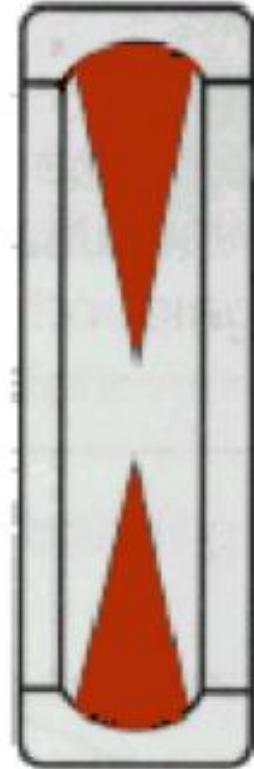
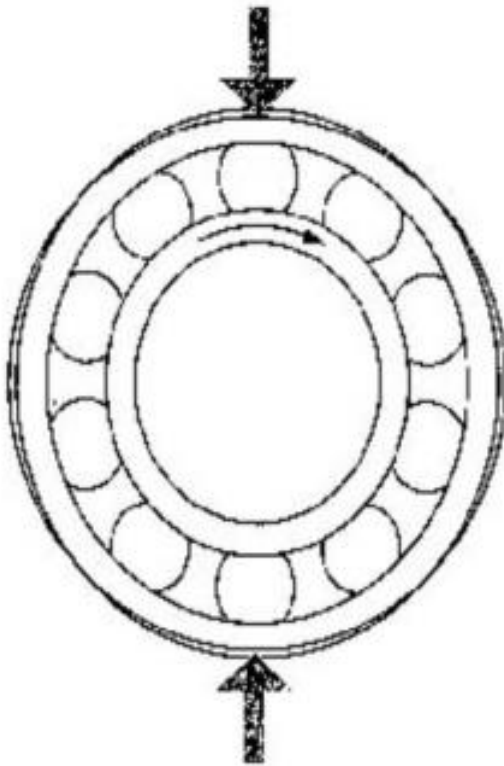
Pure Axial load



Misalignment of outer ring



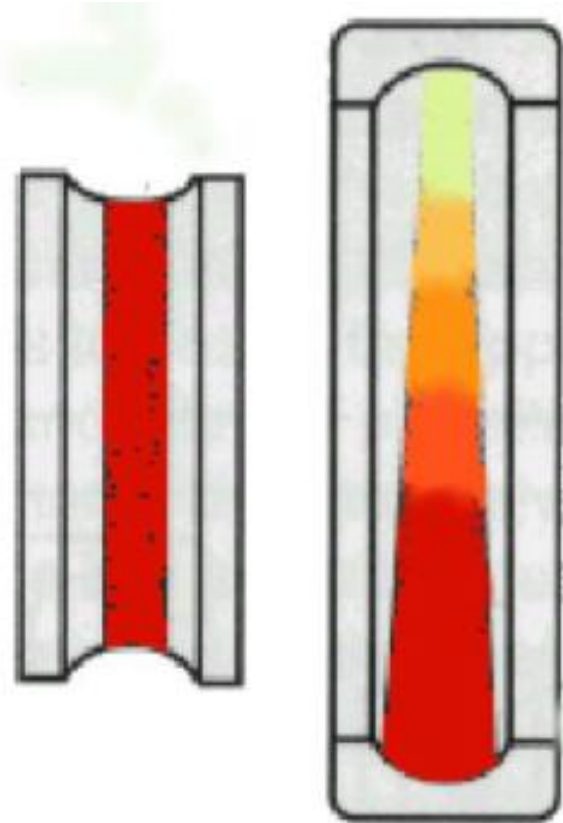
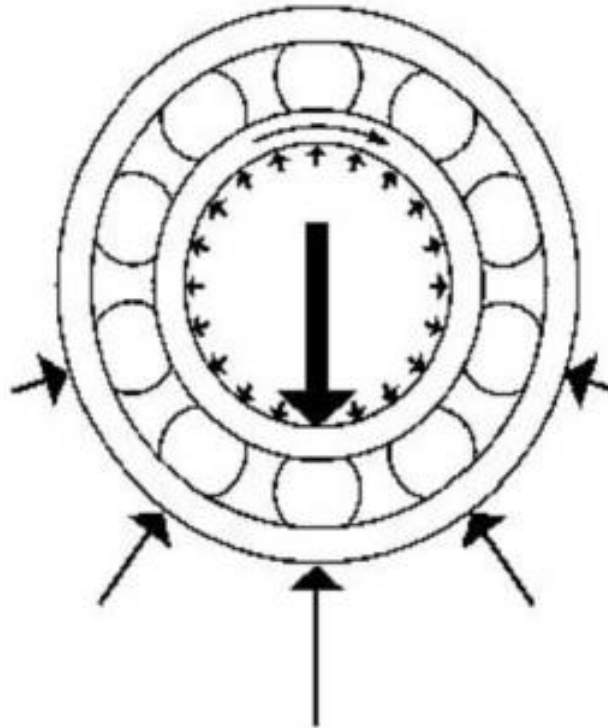
Ovalised outer ring



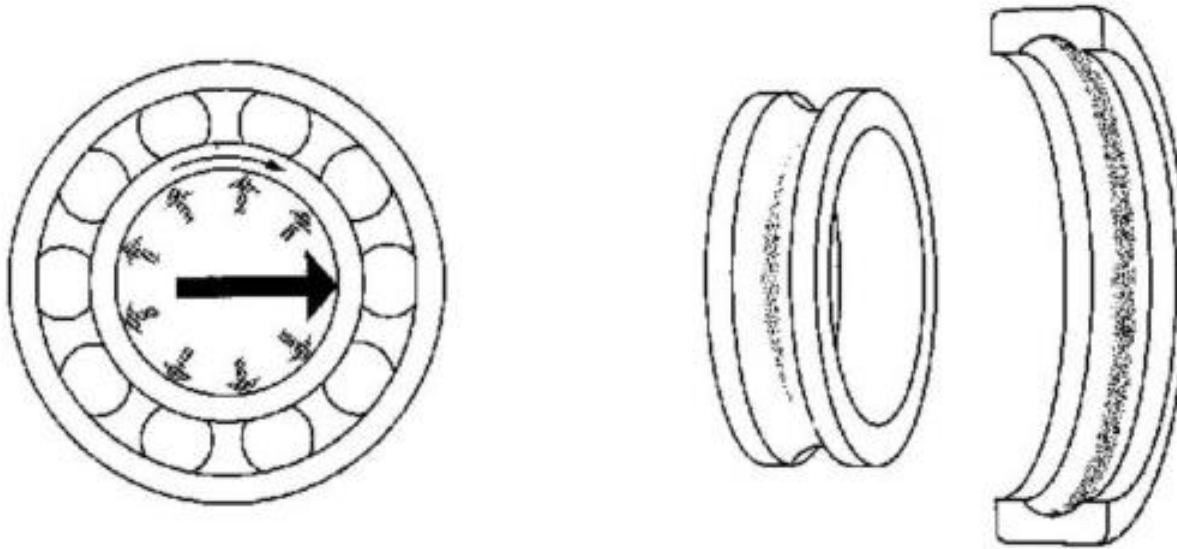
Out of round housing



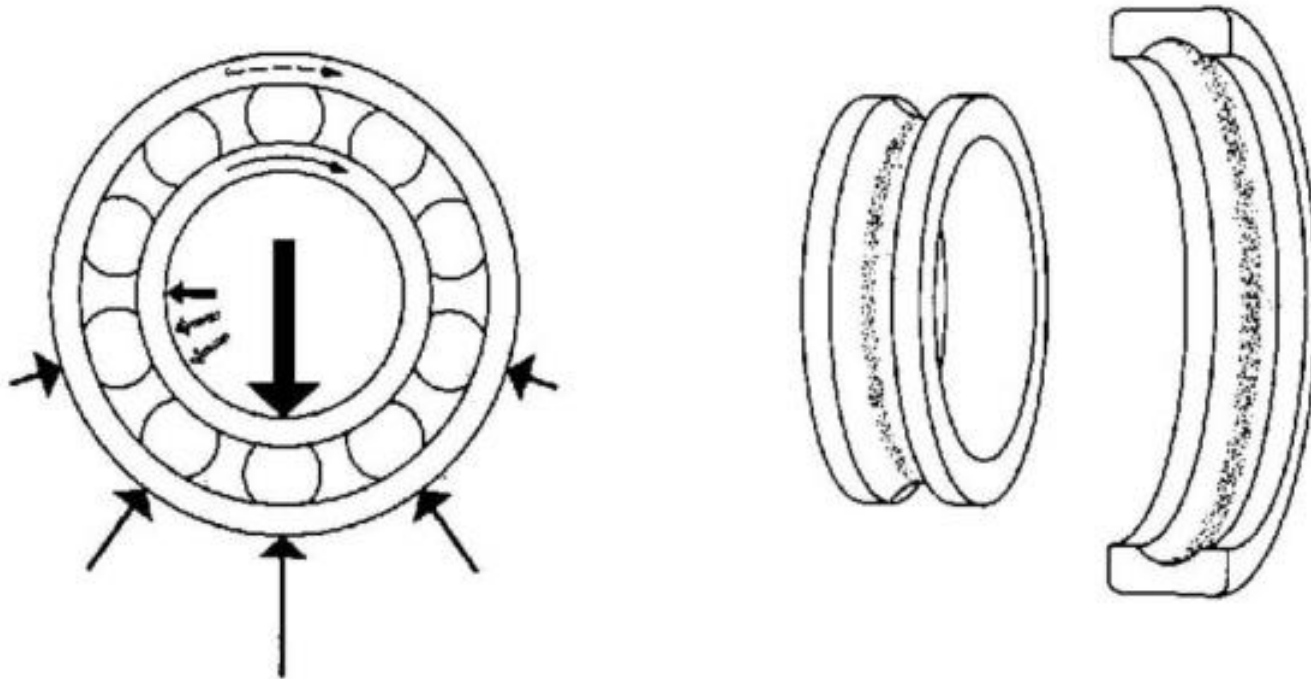
Tight fit - preloading



Eccentric radial load



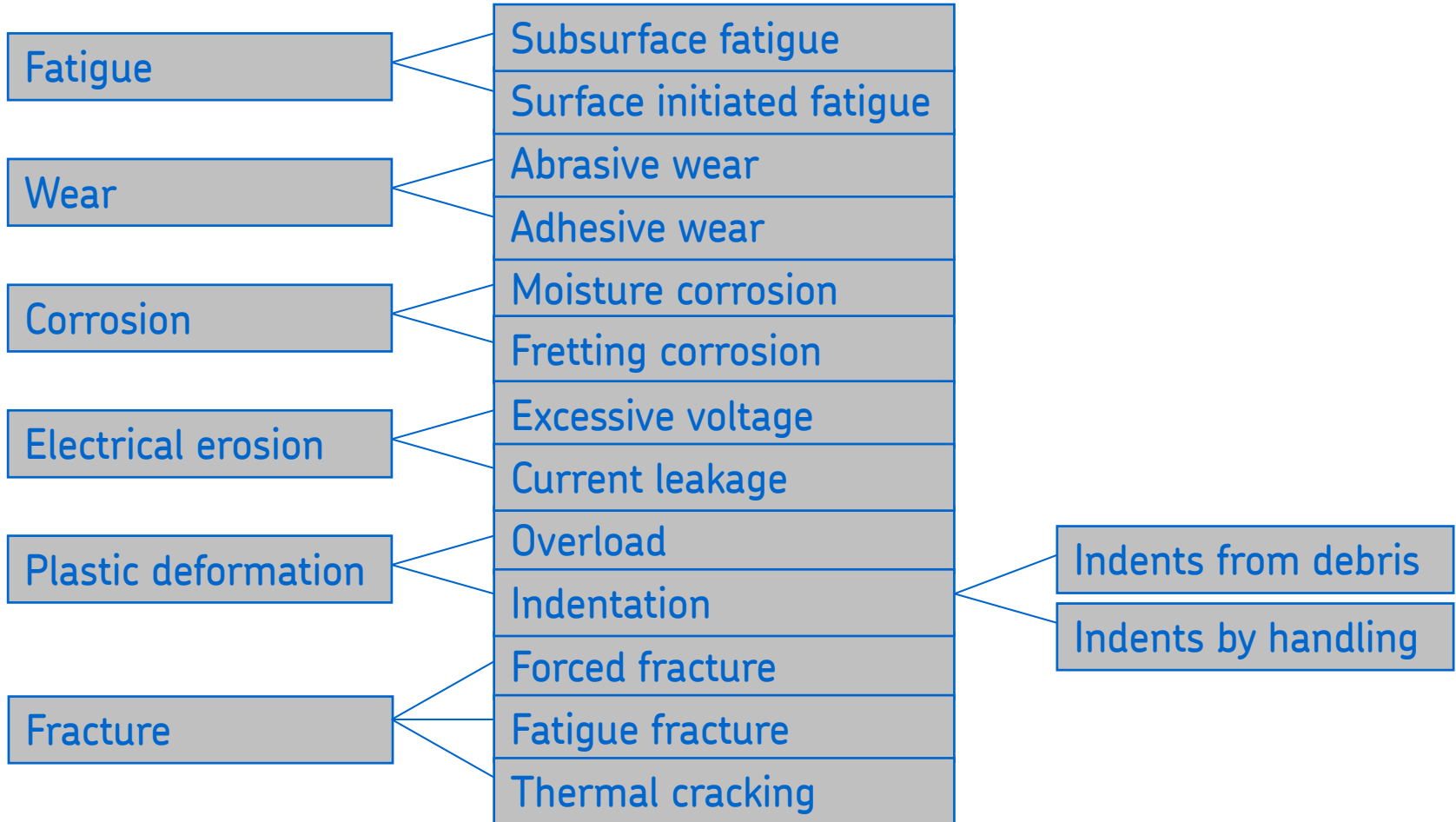
Imbalance



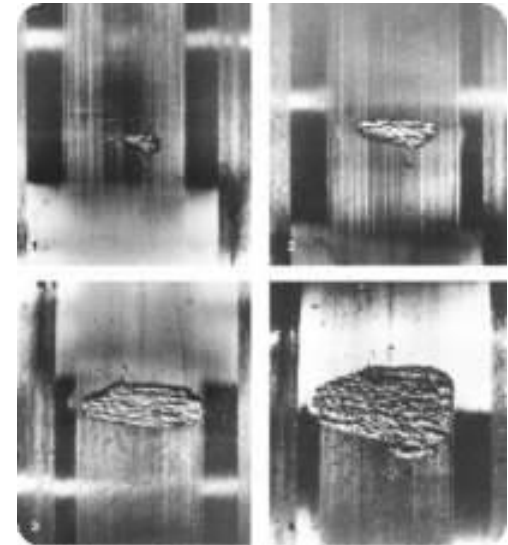
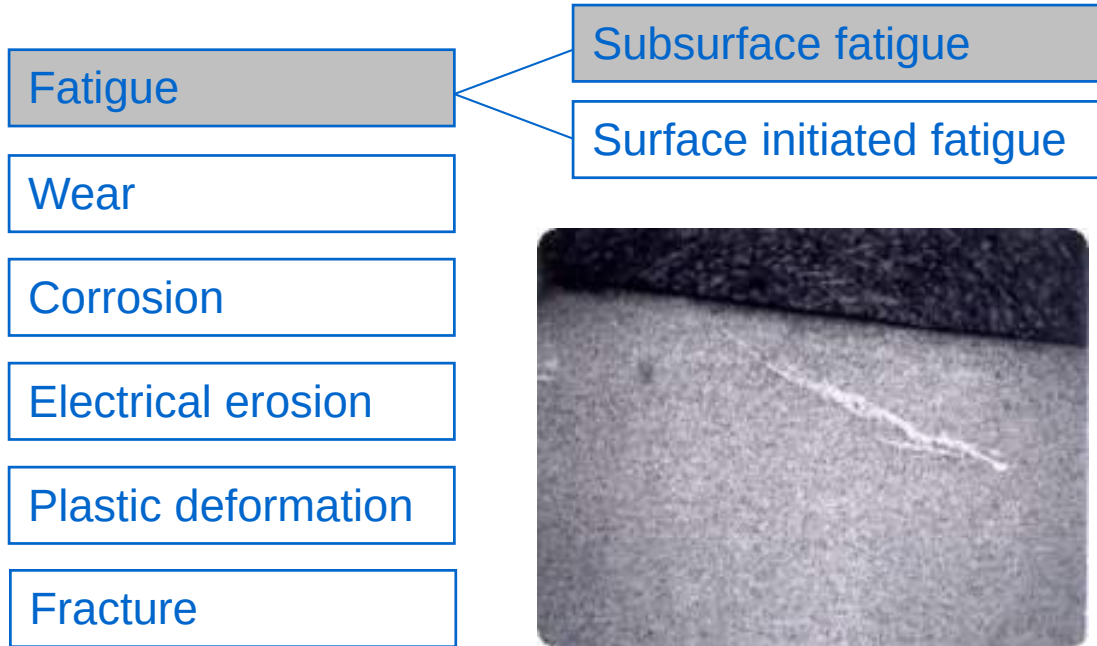
Failure mode classification

- Causes of failures have identifiable characteristics
- Failure mechanisms have identifiable failure modes
- Observed damage can identify failure causes

Failure modes



Subsurface fatigue



- Repeated stress changes
- Material structural
- Micro cracks under the surface changes
- Crack propagation
- Flaking and peeling

Fatigue spall



Edge loading



True brinelling



Surface initiated fatigue

Fatigue

Wear

Corrosion

Electrical erosion

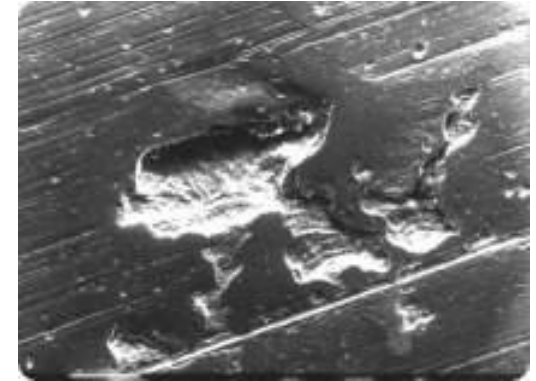
Plastic deformation

Fracture

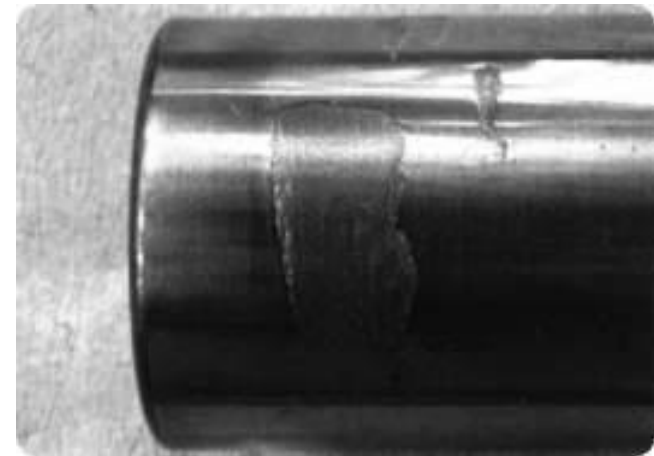
Subsurface fatigue

Surface initiated fatigue

- Surface distress
- Reduced lubrication
- Sliding motion
- Severity micro cracks

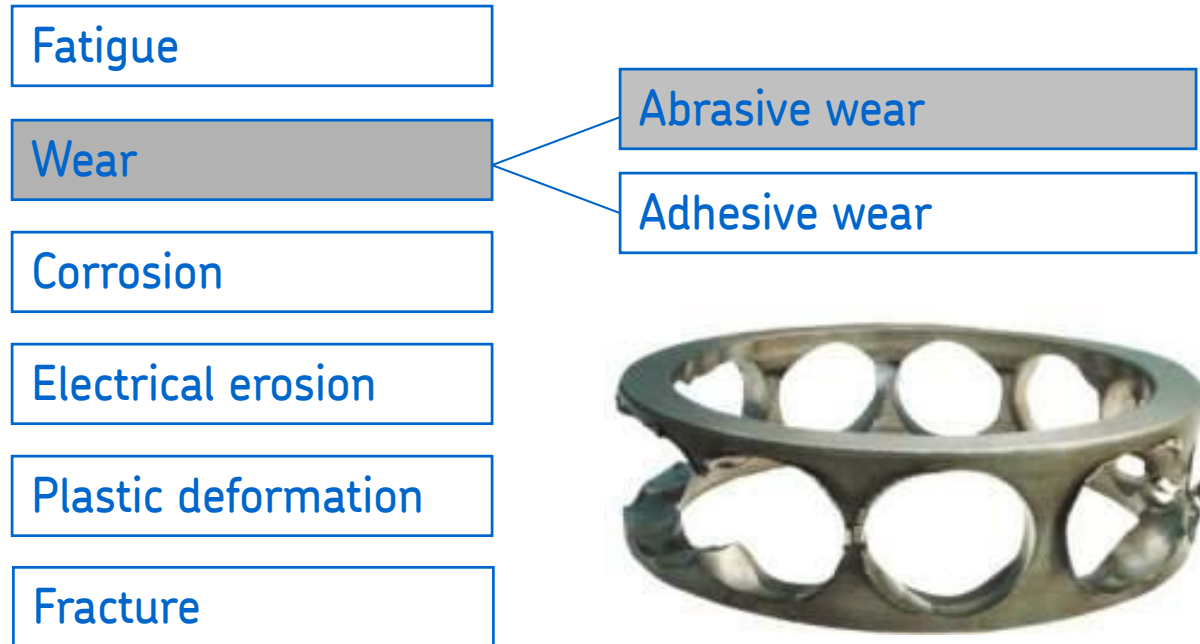


40 μm

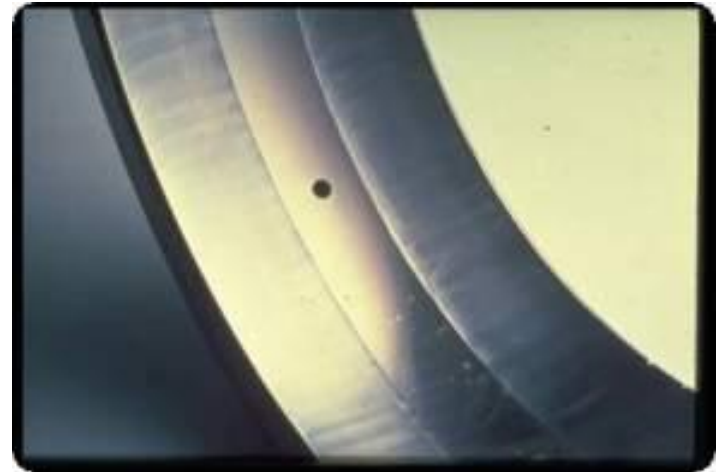


Abrasive wear

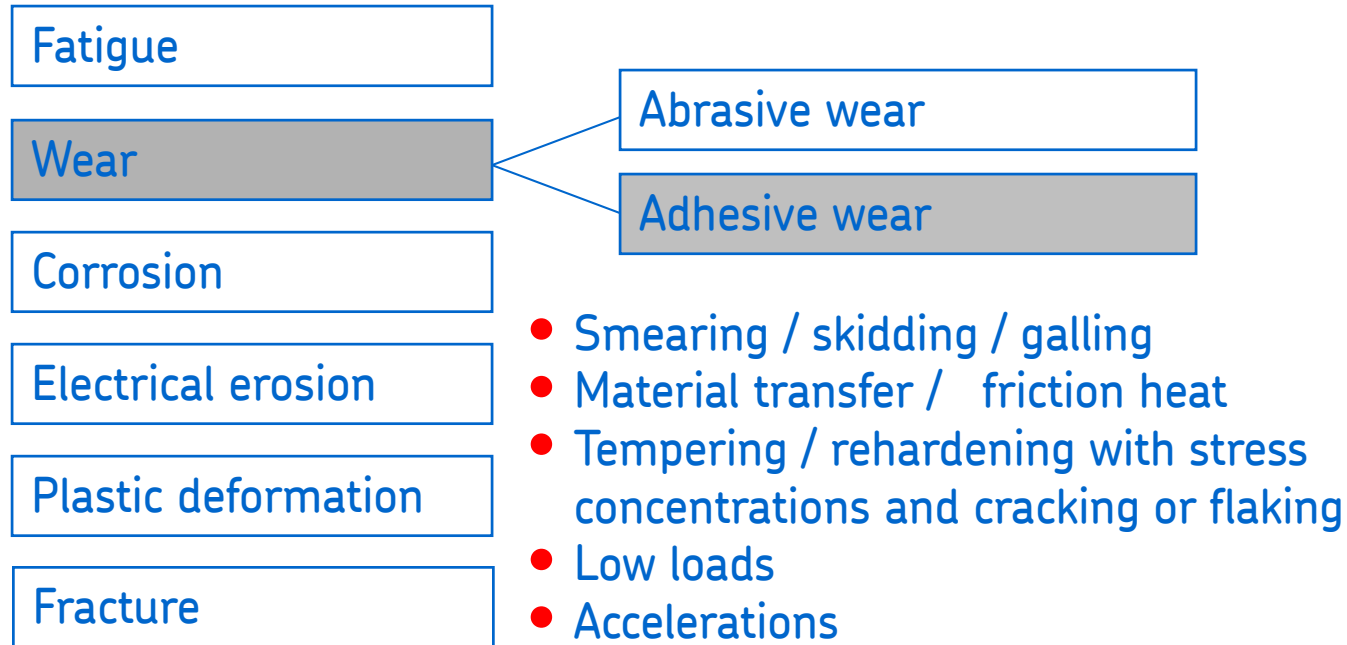
- Progressive removal of material
- Accelerating process
- Inadequate lubrication
- Ingress of dirt particles



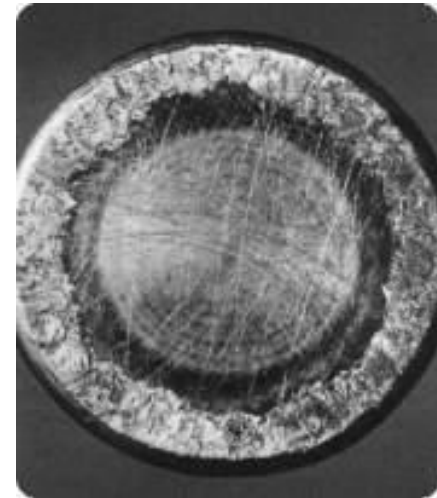
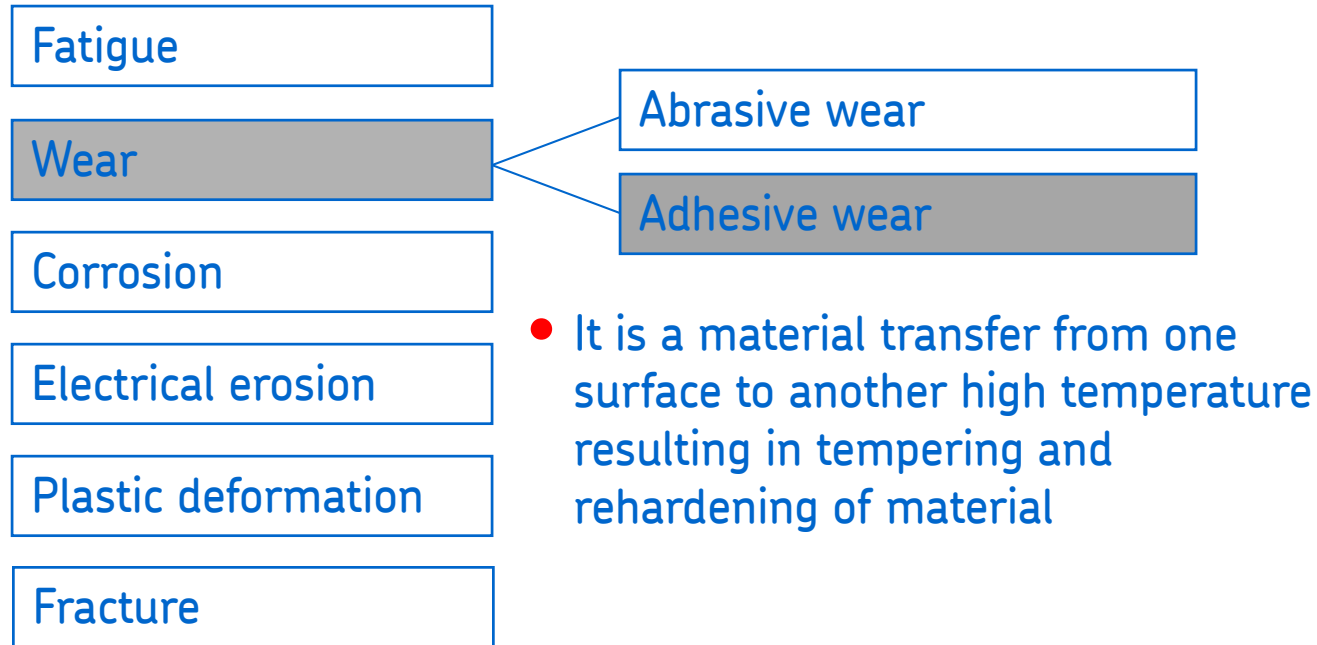
Wear



Adhesive wear



Adhesive wear



Temperature discoloration

	150° - 177° C (300° - 350° F)
	177° - 205° C (350° - 400° F)
	205° - 260° C (400° - 500° F)
	+ 260° C (+ 500° F)
	+ 540° C (+ 1000° F)



- SKF Bearings can be used at temperatures up to 125° C (~ 250° F)
- Higher temperatures may cause loss of Hardness
- Loss of 2-4 points of Rockwell Hardness reduces life 50%

Corrosion

Fatigue

Wear

Corrosion

Electrical erosion

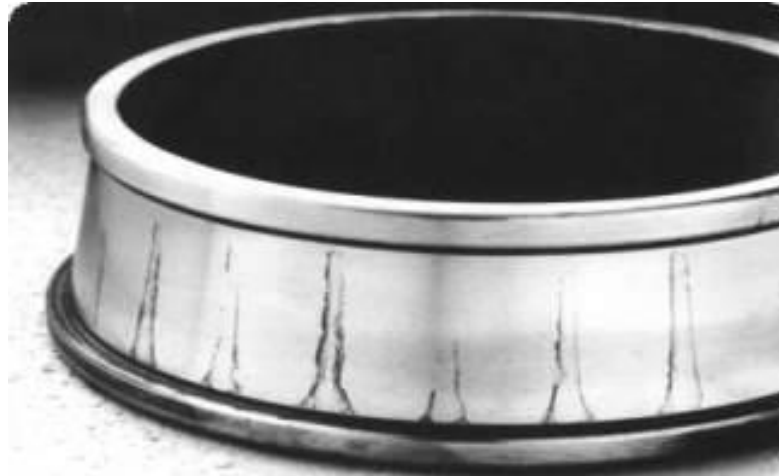
Plastic deformation

Fracture

Rust will form if water or corrosive agents reach the inside of the bearing. In such quantities that the lubricant can not provide adequate protection.

Moisture corrosion

Fretting corrosion

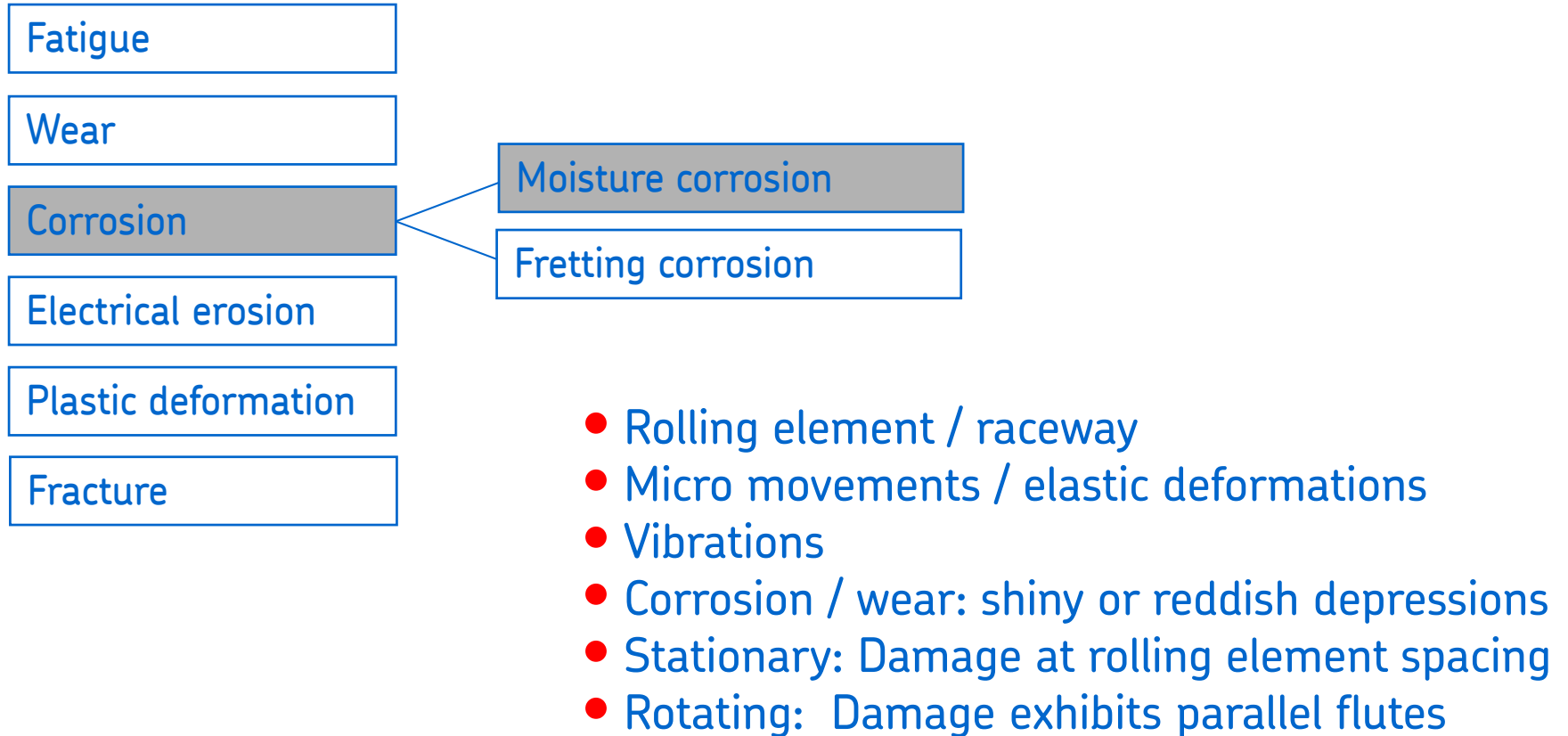


- Oxidation / rust
- Chemical reaction
- Etching (water / oil mixture)

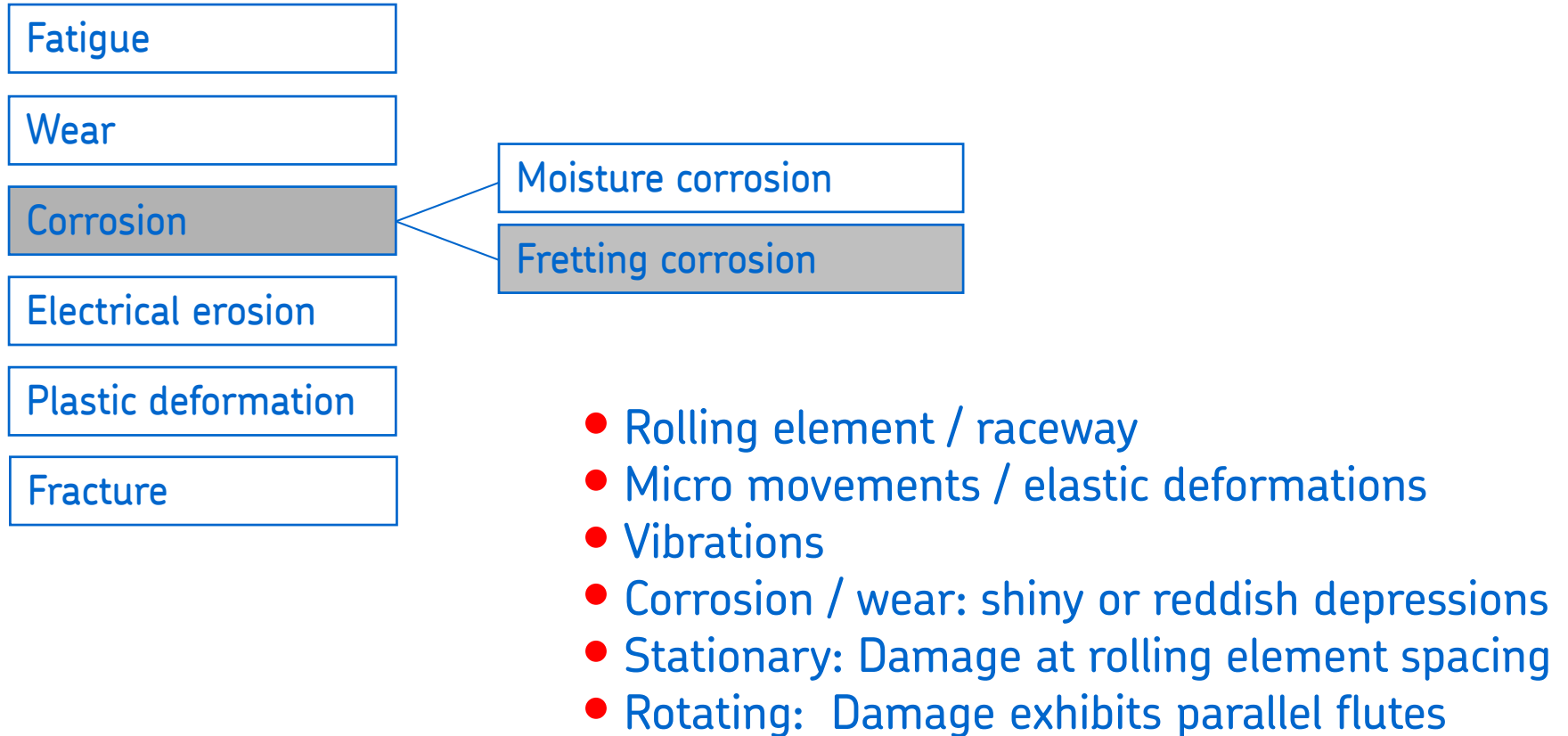
Ineffective sealing



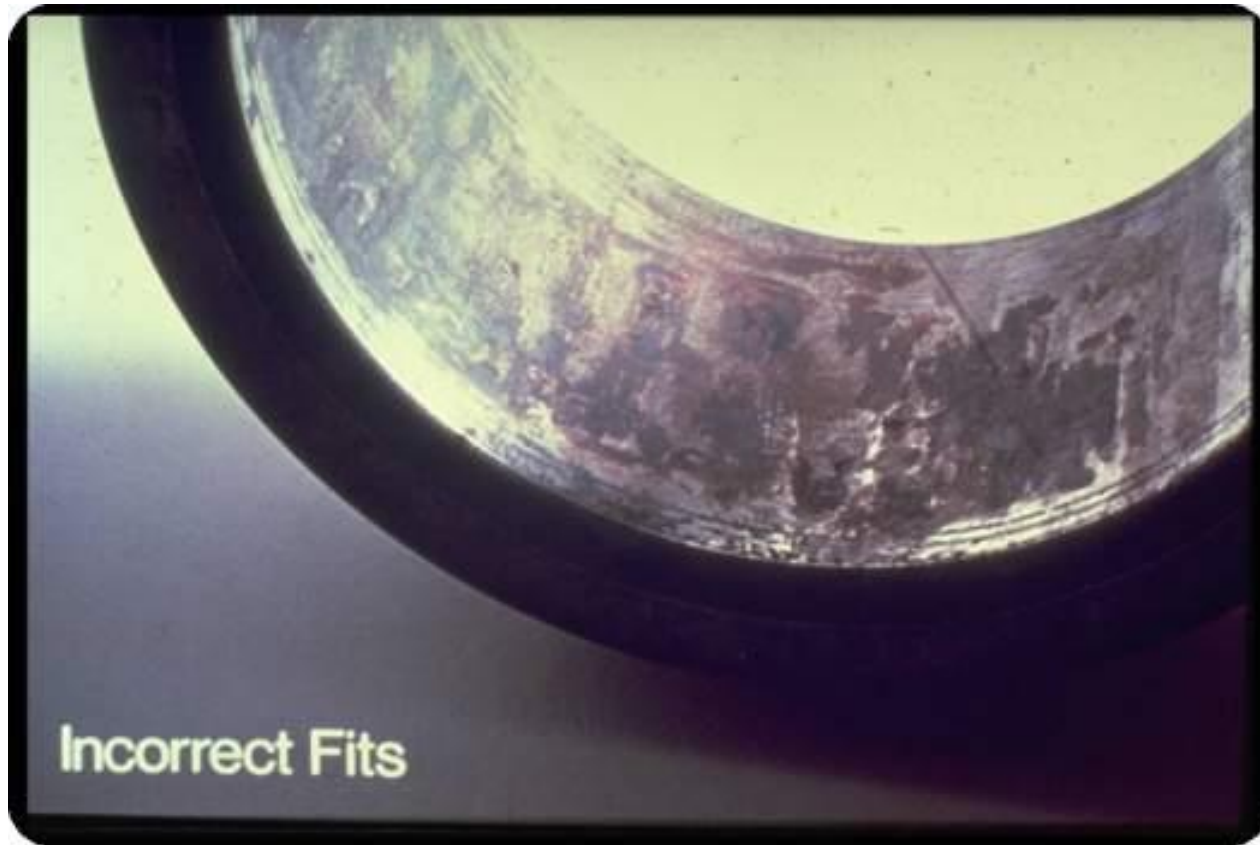
False brinelling...



Fretting corrosion



Advanced fretting...



Advanced fretting



Electrical erosion

- High current = sparking
- Instant localized heating leads to melting and/or welding
- Craters up to 100 μm

Fatigue

Wear

Corrosion

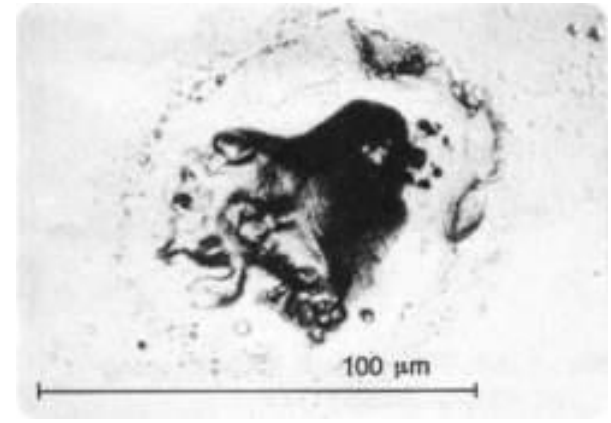
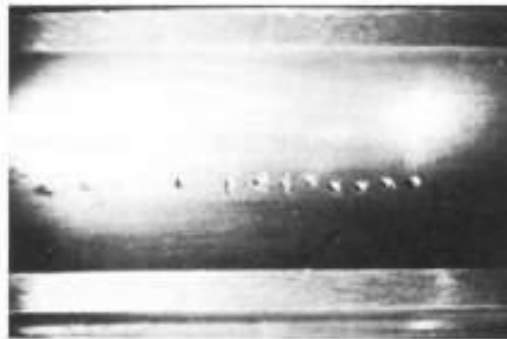
Electrical erosion

Plastic deformation

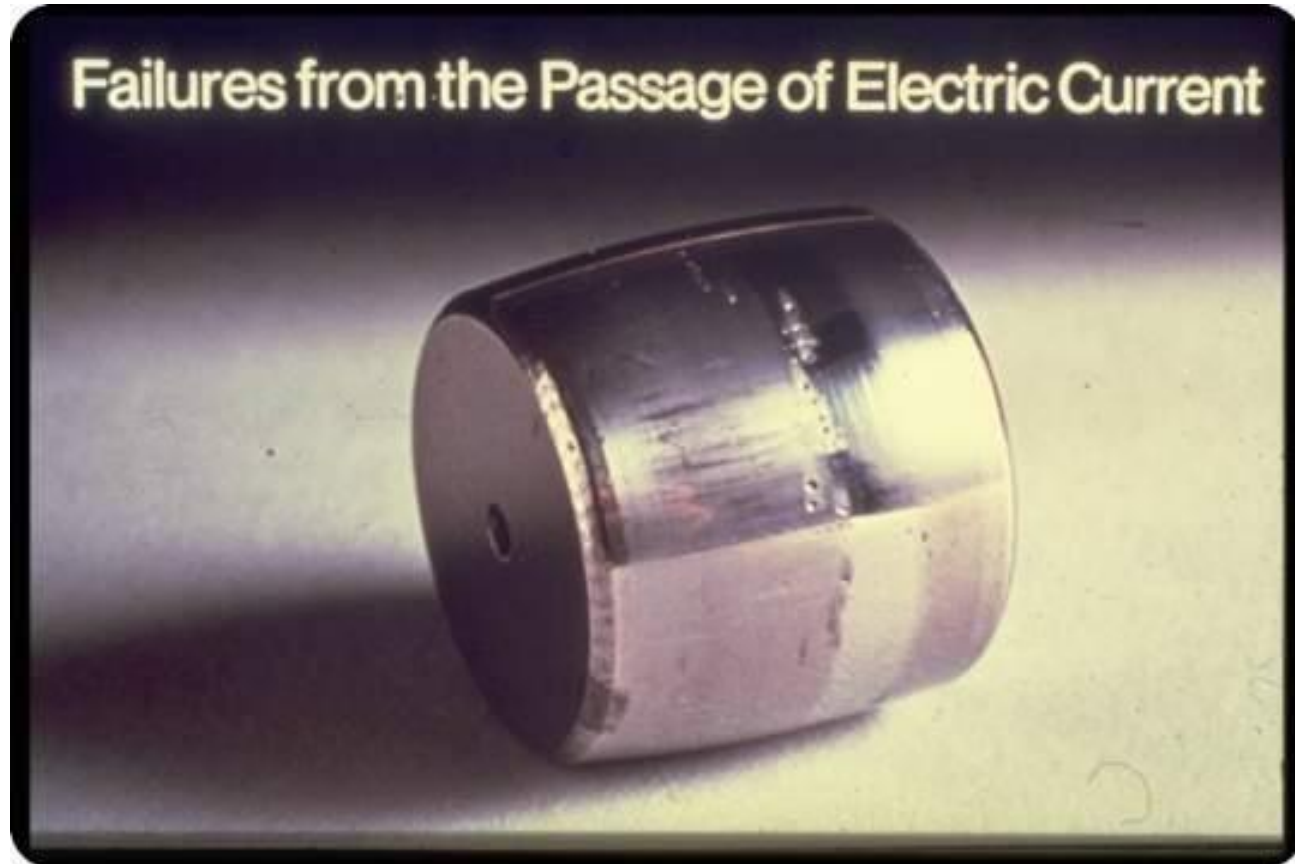
Fracture

Excessive voltage

Current leakage



Electrical erosion – excessive voltage



Electrical erosion

Fatigue

Wear

Corrosion

Electrical erosion

Plastic deformation

Fracture

Excessive voltage

Current leakage

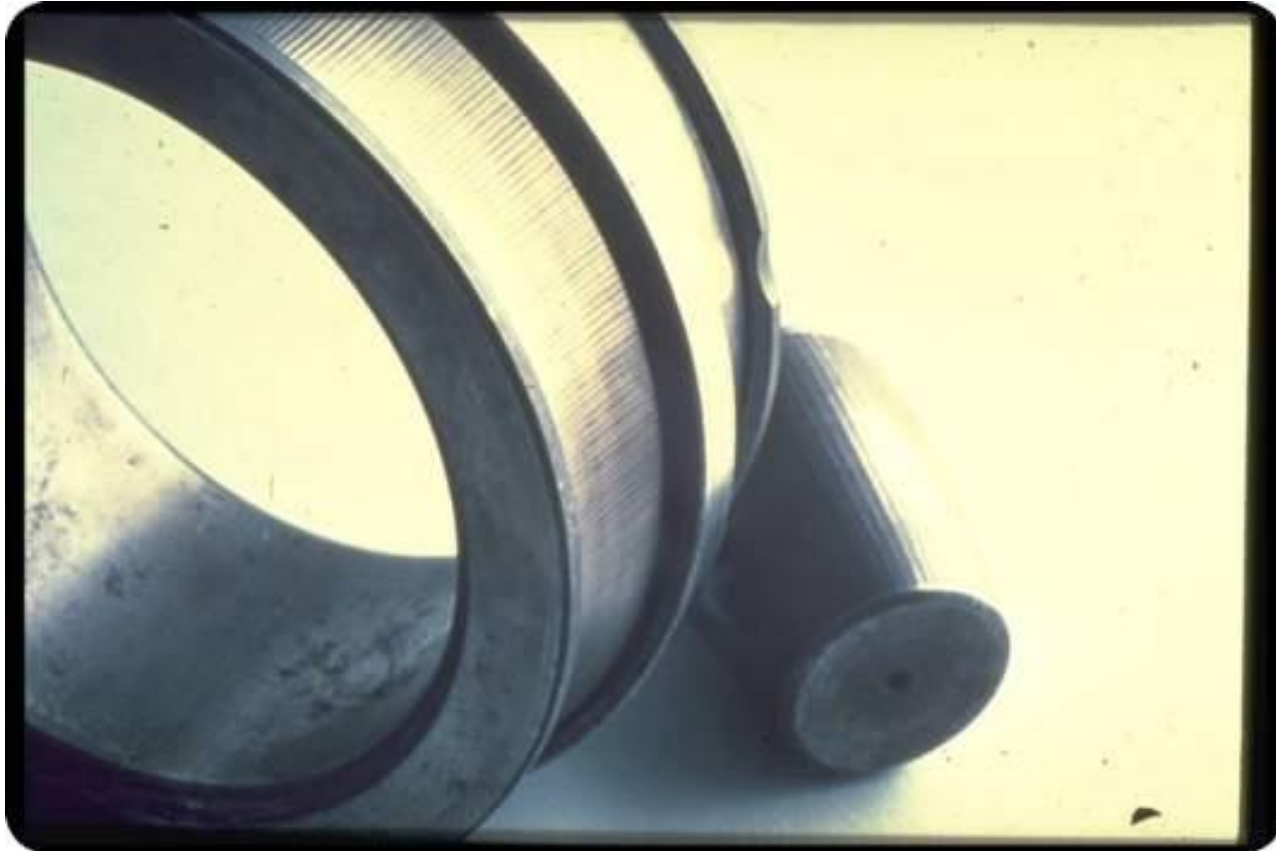
- Low current intensity
- Shallow craters closely positioned
- Development of flutes on raceways and rollers, parallel to rolling axis
- Dark gray discoloration



Vibration



Current leakage



Electric current passage solutions



Hybrid deep groove
ball bearing



Insocoat

Overload

Fatigue

Wear

Corrosion

Electrical erosion

Plastic deformation

Fracture

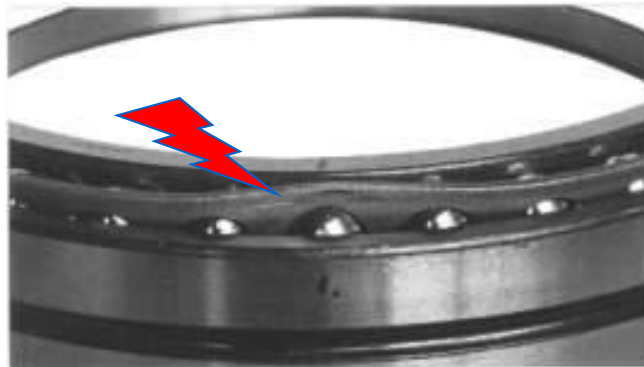
- Static or shock loads
- Plastic deformations
- Depressions at rolling element spacing
- Handling damage

Overload

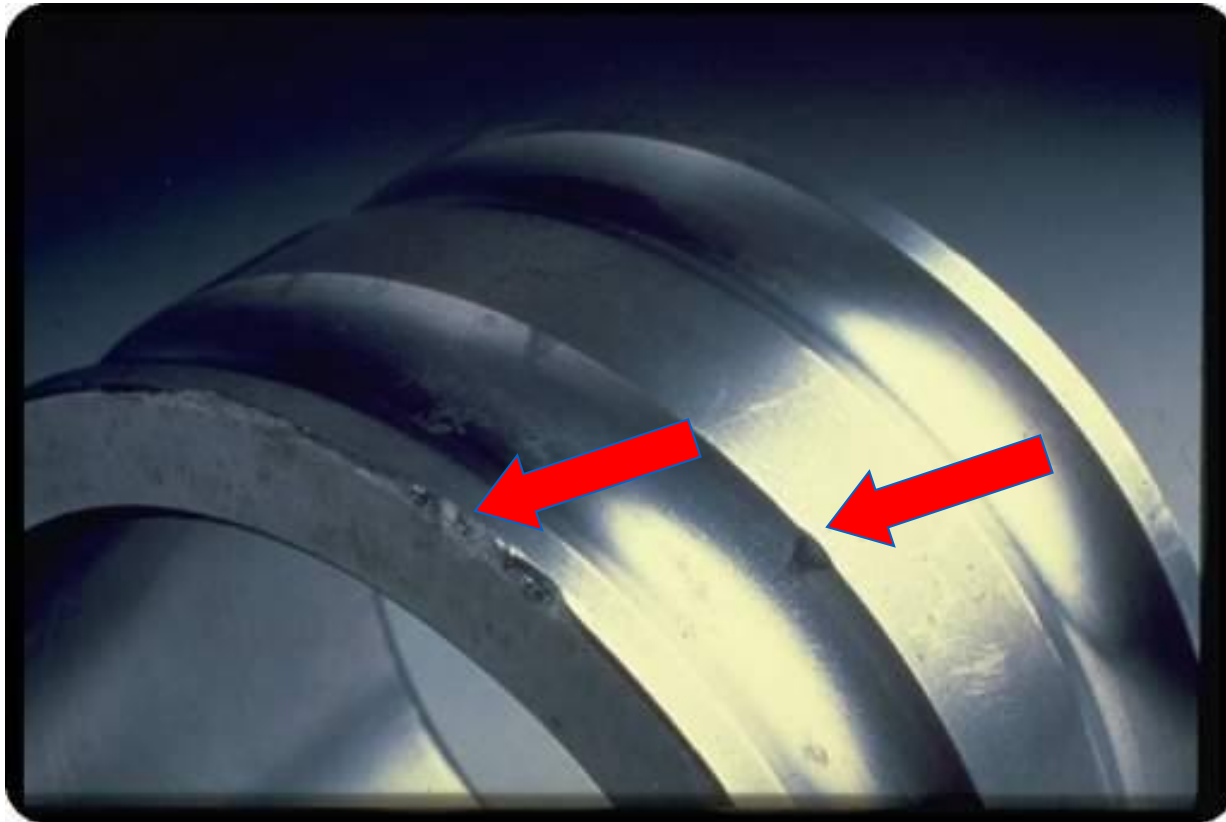
Indentation

Indents from debris

Indents by handling



Installation damage



Debris denting

Fatigue

Wear

Corrosion

Electrical erosion

Plastic deformation

Fracture

- Localized overloading
- Over-rolling of particles = dents
- Caused by soft / hardened steel / hard mineral particles

Overload

Indentation

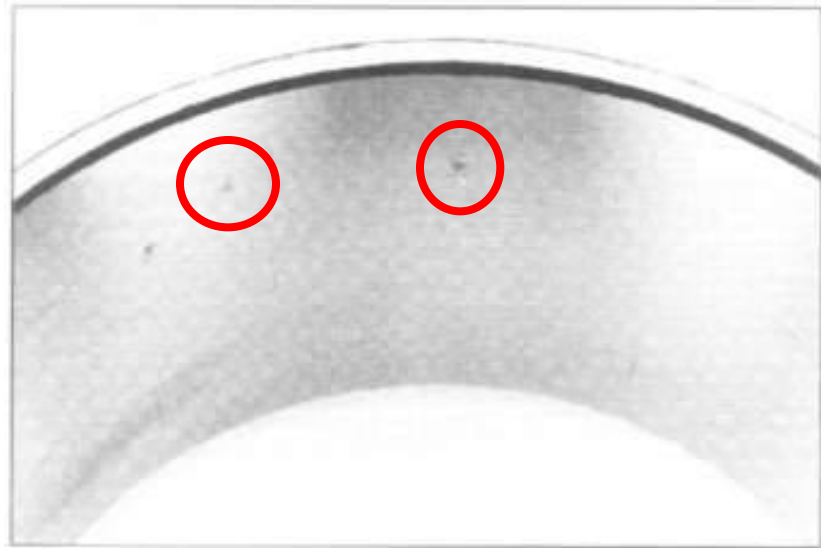
Indents from debris

Indents by handling



Handling damage...

- Localized overloading
- Nicks caused by hard / sharp objects



Fatigue

Wear

Corrosion

Electrical erosion

Plastic deformation

Fracture

Overload

Indentation

Indents from debris

Indents by handling

Handling damage



CRB roller damage



Forced fracture...

Fatigue

Wear

Corrosion

Electrical erosion

Plastic deformation

Fracture

- Stress concentration exceeds tensile strength
- Impact / overstressing

Forced fracture

Fatigue fracture

Thermal cracking



Forced fracture...



Fatigue fracture

Fatigue

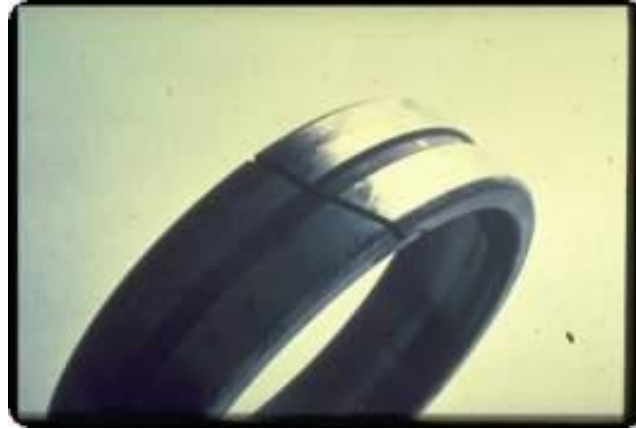
Wear

Corrosion

Electrical erosion

Plastic deformation

Fracture



Forced fracture

Fatigue fracture

Thermal cracking

- Exceeding fatigue strength under bending
- Crack initiation / propagation
- Finally forced fracture
- Rings and Cages

Thermal cracking

Fatigue

Wear

Corrosion

Electrical erosion

Plastic deformation

Fracture

- High sliding and / or insufficient lubrication
- High friction heat
- Cracks at right angle to sliding direction

Forced fracture

Fatigue fracture

Thermal cracking



Securing evidence

- Collect operating data, monitoring data
- Collect lubricant samples
- Check bearing environment(s)
- Assess bearing(s) in mounted condition
- Mark mounting position(s)
- Remove, mark and bag bearing(s) and parts
- Check bearing seats

Conducting the analysis

Examine Bearing(s) and Parts

- Record visual observations
- Use the Failure Modes to eliminate improbable possible causes and determine the original cause of the failure.
- Contact external resources for assistance, if needed
- Initiate corrective action, if desired

False brinelling

